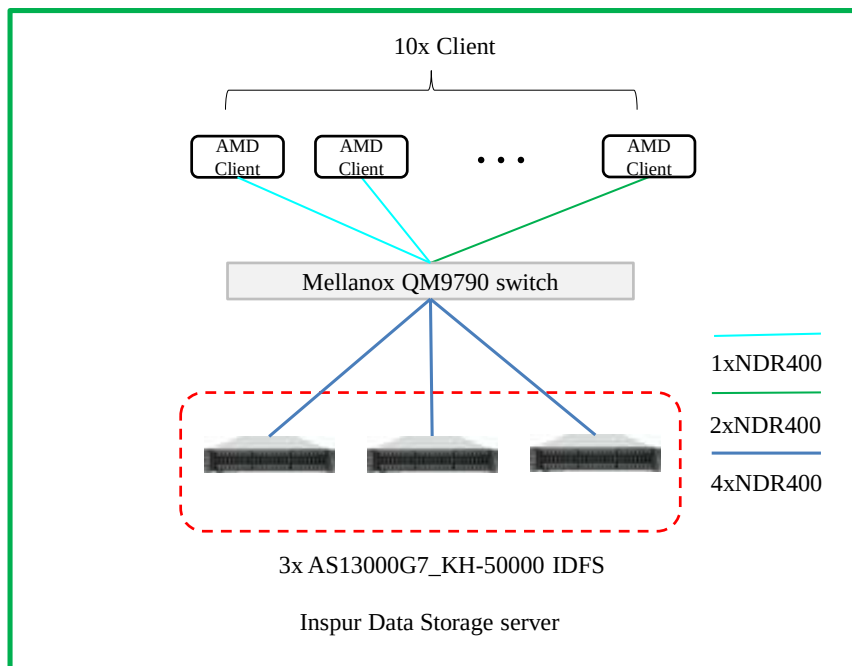

1. Overview

This benchmark test uses a 3-node storage server cluster running the AS13000G7 IDFS distributed storage system.

2. Hardware Diagram



2.1 Client

This test utilized a total of 10 client nodes with three distinct hardware configurations as follows:

4 client nodes: Equipped with 1 AMD 9355 CPU, 768 GB memory, two 400 Gb IB network adapters, and one 1600 W power supply module each.

3 client nodes: Equipped with 1 AMD 9554 CPU, 384 GB memory, two 400 Gb IB network adapter, and one 1600 W power supply module each.

1 client nodes: Equipped with 1 AMD 9554 CPU, 512 GB memory, two 400 Gb IB network adapter, and one 1600 W power supply module each.

1 client nodes: Equipped with 1 AMD 9554 CPU, 512 GB memory, one 400 Gb IB network adapter, and one 1600 W power supply module each.

1 client nodes: Equipped with 1 AMD 9554 CPU, 384 GB memory, one 400 Gb IB network adapter, and one 1600 W power supply module each.

2.2 Server

The storage cluster is composed of three 2U rack-mount AS13000G7_KH-50000 storage appliances. Each node is physically populated with two KH-50000 CPUs, sixteen 7.68TiB NVMe SSDs and four 400Gb InfiniBand adapters, with the IDFS distributed file system deployed across the cluster.



2.3 Server Power Requirements

System component	Power supply unit	Nameplate rated power	Design Power
AS13000G7_KH-50000 Appliance 1	Power supply 1	1600W	1600W
	Power supply 2	1600W	
Totals		3200W	1600W

2.4 Switch

Mellanox QM9790 switch.

3. Software

3.1 Client

Client OS – Ubuntu 22.04

Mellanox OFED version – 5.8-4.1.5.0

3.2 Server

Storage Servers OS – Alibaba Cloud Linux release 3

Storage Servers Kernel – Linux icfs41 5.10.134-010.ali5000.pro.a18.x86_64

Mellanox OFED version – 4.10.0.6.6.0

Storage Servers version – idfs version 7.2